

Part imagery & 3D — accurate, legal, offline (your call)

Goal: replace vague 3D blobs with the EXACT correlative image for each part, and show it in the collection preview boxes.

Hard caveats — why 'download real CAD for every part' isn't the path

ITAR: CAD/repair/maintenance data for military NSN parts is CONTROLLED technical data — not freely downloadable, not bundleable, and this is an OFFLINE system anyway.

Web CAD (GrabCAD/TraceParts/McMaster): license-restricted, can't be redistributed, and rarely matches the exact NSN/variant.

AI-generated 3D (text/image -> mesh): produces an APPROXIMATION, not engineering-accurate geometry. Presenting it as authoritative in a maintenance tool is a safety/accuracy risk.

So 'find/scan/generate accurate CAD for all parts' can't be done safely or legally. But the GOAL — an accurate, correlative image per part — is very achievable:

The accurate, legal, offline answer

☐ Use the manual's OWN figure

- Every part already cites its figure + page (parts table).
- Crop that exploded-view illustration -> the EXACT, cited image of that part.
- US Army TMs are public-domain; nothing to license.

Authoritative & correlative.

☐ Preview boxes = real results

- The 3D collection cards show that crop (the actual part),
- not a generic blob. Modal gets a 'Manual illustration' tab.
- Lazy-extract + cache as a sidecar (index never written).

Exactly what you asked for.

☐ Keep parametric 3D (honest)

- For primitive families (bolt/nut/gasket/bearing) it's
- dimensionally accurate from FLIS — keep as the interactive
- view, clearly labeled 'representative', paired with the crop.

Interactive where it's truthful.

GPU's real role here (honest)

GPU genuinely helps: batch figure-region + caption detection across the corpus, and an OPTIONAL offline image-similarity pass to pick the best crop per part. But the PDF page render itself is mostly CPU/IO — so this is an accuracy win, not primarily a GPU-horsepower win. I'll be straight about that.

Recommended (pick any mix — your final call)

A) Figure-crop extraction = authoritative image in preview boxes + a 'Manual illustration' tab (DEFAULT, recommended). B) Keep/strengthen FLIS-driven parametric 3D as the interactive 'representative' view. C) OPT-IN only, clearly labeled offline approximation for parts with no figure — I recommend NOT making this default in a maintenance tool. All sidecar/cached, RPS-safe, index never written.